Year 10 Mathematics

Algebra 2

7.2 Simplifying Algebraic Fractions

Simplifying Algebraic Fractions

STARTER · 5 MIN

Date: _____ **Title:** Simplifying Algebraic Fractions

LEARNING OBJECTIVES

- Simplify algebraic fractions by cancelling common factors
- Factorise to simplify algebraic fractions
- Multiply and divide algebraic fractions

STARTER *Answer in your book. No calculators.*

1. Simplify $\frac{12}{18}$

2. Factorise $3x + 6$

3. Factorise $5y - 10$

4. Simplify $x^5 \div x^2$

Starter — Answers

STARTER

Tick or correct in green pen

1. $\frac{12}{18} = \frac{\mathbf{2}}{\mathbf{3}}$

2. $3x + 6 = \mathbf{3(x + 2)}$

3. $5y - 10 = \mathbf{5(y - 2)}$

4. $x^{5-2} = \mathbf{x^3}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can simplify algebraic fractions by cancelling common factors	☐
SILVER	I can factorise to simplify algebraic fractions	☐
GOLD	I can multiply and divide algebraic fractions	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Algebraic Fractions

INTRODUCTION · 3 MIN

Simplify algebraic fractions the same way as number fractions: cancel common factors.

Numbers

Divide top and bottom by the HCF.

$$\frac{6x}{9} = \frac{2x}{3}$$

Letters

Subtract the indices.

$$\frac{x^5}{x^2} = x^3$$

Factorise first

Only cancel whole factors.

$$\frac{2x+4}{6} = \frac{2(x+2)}{6} = \frac{x+2}{3}$$

Multiply

Multiply tops and bottoms, then cancel.

$$\frac{x}{2} \times \frac{4}{x^2} = \frac{2}{x}$$

Examples 1 and 2 — Cancelling Common Factors

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Simplify

$$\frac{8x}{12}$$

Example 2

Simplify

$$\frac{10a^4}{15a}$$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$\frac{8x}{12} = \frac{8x \div 4}{12 \div 4}$$

$$= \mathbf{\frac{2x}{3}}$$

Tip: find the HCF of the numbers.

Example 2

$$\frac{10a^4}{15a} = \frac{2a^{4-1}}{3}$$

$$= \mathbf{\frac{2a^3}{3}}$$

Tip: cancel the numbers, then the letters.

Task 1

PRACTICE · 9 MIN

BRONZE

Simplify each fraction

1. $\frac{6x}{9}$

2. $\frac{4y}{12}$

3. $\frac{x^5}{x^2}$

4. $\frac{12a^3}{4a}$

5. $\frac{15p^2q}{5pq}$

★ $\frac{9x^2y^3}{6xy}$

Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1 $\frac{6x}{9} = \frac{\mathbf{2x}}{\mathbf{3}}$

2 $\frac{4y}{12} = \frac{\mathbf{y}}{\mathbf{3}}$

3 $\frac{x^5}{x^2} = \mathbf{x^3}$

4 $\frac{12a^3}{4a} = \mathbf{3a^2}$

5 $\frac{15p^2q}{5pq} = \mathbf{3p}$

★ $\frac{9x^2y^3}{6xy} = \frac{\mathbf{3xy^2}}{\mathbf{2}}$

Examples 3 and 4 — Factorising First

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $a^2 - b^2 = (a + b)(a - b)$

Example 3

Simplify

$$\frac{3x + 9}{6}$$

Example 4

Simplify

$$\frac{4x - 8}{x - 2}$$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$\frac{3x+9}{6} = \frac{3(x+3)}{6}$$

$$= \frac{\mathbf{x+3}}{\mathbf{2}}$$

Tip: factorise the top, then cancel.

Example 4

$$\frac{4x-8}{x-2} = \frac{4(x-2)}{x-2}$$

$$\mathbf{= 4}$$

Tip: cancel the whole bracket.

Task 2

PRACTICE · 9 MIN

SILVER *Simplify each fraction*

Hint $a^2 - b^2 = (a + b)(a - b)$

1. $\dfrac{2x + 6}{4}$

2. $\dfrac{5y - 15}{10}$

3. $\dfrac{6a + 12}{a + 2}$

4. $\dfrac{x^2 + 3x}{x}$

5. $\dfrac{3x + 3}{6x + 6}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $\frac{2(x+3)}{4} = \frac{\mathbf{x+3}}{\mathbf{2}}$

2 $\frac{5(y-3)}{10} = \frac{\mathbf{y-3}}{\mathbf{2}}$

3 $\frac{6(a+2)}{a+2} = \mathbf{6}$

4 $\frac{x(x+3)}{x} = \mathbf{x+3}$

5 $\frac{3(x+1)}{6(x+1)} = \frac{\mathbf{1}}{\mathbf{2}}$

Examples 5 and 6 — Multiplying and Dividing

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $a^2 - b^2 = (a + b)(a - b)$

Example 5

(2 marks)

Simplify

$$\frac{2x}{5} \times \frac{10}{x^2}$$

Example 6

(2 marks)

Simplify

$$\frac{3a}{4} \div \frac{9a^2}{8}$$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$\frac{20x}{5x^2} = \mathbf{\frac{4}{x}}$$

Tip: multiply tops and bottoms, then cancel.

Example 6

$$\frac{3a}{4} \times \frac{8}{9a^2} = \frac{24a}{36a^2} = \mathbf{\frac{2}{3a}}$$

Tip: keep, change, flip.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $a^2 - b^2 = (a + b)(a - b)$

1 Simplify $\frac{x}{3} \times \frac{6}{x^2}$ (2 marks)

2 Simplify $\frac{4y^2}{5} \times \frac{15}{2y}$ (2 marks)

3 Simplify $\frac{2x}{7} \div \frac{4x}{21}$ (2 marks)

4 Simplify $\frac{x+1}{3} \times \frac{6}{2x+2}$ (3 marks)

5 Simplify $\frac{5a^3}{2b} \div \frac{10a}{b^2}$ (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $\frac{6x}{3x^2} = \mathbf{\frac{2}{x}}$

2 $\frac{60y^2}{10y} = \mathbf{6y}$

3 $\frac{2x}{7} \times \frac{21}{4x} = \mathbf{\frac{3}{2}}$

4 $\frac{6(x+1)}{6(x+1)} = \mathbf{1}$

5 $\frac{5a^3}{2b} \times \frac{b^2}{10a} = \mathbf{\frac{a^2b}{4}}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $\frac{14x}{21}$

SILVER Simplify $\frac{4x + 12}{8}$

GOLD Simplify $\frac{3x}{4} \times \frac{8}{9x}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE

$$\frac{14x}{21} = \frac{2x}{3}$$

SILVER

$$\frac{4(x+3)}{8} = \frac{x+3}{2}$$

GOLD

$$\frac{24x}{36x} = \frac{2}{3}$$

Exam Practice — Cancelling Common Factors

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

Simplify fully

(a) (1)

$$\frac{6x}{9}$$

.....

(b) (1)

$$\frac{4ab}{10a^2}$$

.....

Question 2

(2 marks)

Simplify fully

$$\frac{15x^2y}{5xy^3}$$

.....

Exam Practice — Factorising First

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $a^2 - b^2 = (a + b)(a - b)$

Question 3

(3 marks)

Simplify fully

(a) (1)

$$\frac{3x + 6}{x + 2}$$

.....

(b) (2)

$$\frac{x^2 + 5x}{x}$$

.....

Question 4

(3 marks)

Simplify fully

$$\frac{x^2 - 9}{x^2 + 5x + 6}$$

.....

Exam Practice — Multiplying and Dividing

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $a^2 - b^2 = (a + b)(a - b)$

Question 5

(2 marks)

Simplify fully

$$\frac{2x}{3} \times \frac{9}{4x^2}$$

.....

Question 6

(3 marks)

Simplify fully

$$\frac{x + 3}{4} \div \frac{x^2 - 9}{8}$$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $\frac{2x}{3}$ — B1

(b) $\frac{2b}{5a}$ — B1

Tip: cancel numbers first, then letters.

Question 2

$15 \div 5 = 3$, $x^2 \div x = x$ — M1

$\frac{3x}{y^2}$ — A1

Tip: $y \div y^3$ leaves y^2 on the bottom.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $\dfrac{3(x+2)}{x+2} = 3$ — **B1**

(b) $\dfrac{x(x+5)}{x}$ — **M1**

$= x + 5$ — **A1**

Tip: factorise, then cancel whole brackets.

Question 4

$\dfrac{(x+3)(x-3)}{(x+2)(x+3)}$ — **M1 M1**

$= \dfrac{x-3}{x+2}$ — **A1**

Tip: $x^2 - 9$ is a difference of two squares.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$$\frac{18x}{12x^2}$$ M1

$$= \frac{3}{2x}$$ A1

Tip: multiply tops, multiply bottoms, then cancel.

Question 6

$$\frac{x+3}{4} \times \frac{8}{(x+3)(x-3)}$$ M1 M1

$$= \frac{2}{x-3}$$ A1

Tip: ÷ a fraction = × its reciprocal.